



TIPS & TRICKS



Use hashtags to simplify searching for commands

We use hashtags all over in Twitter. They are excellent for discovering and tagging long text. A # in the shell is used to comment a line. Hashtags can also make our workflow more productive. Let's see how this happens.

We normally type very long commands in our Bash shell and it becomes difficult to search for the same command from History. Here is a simple trick that will make searching easier. All you need to do is to append a hashtag at the end of every command you type. And later search for the hashtag in *reverse-i-search* (*Ctrl + R*). Adding a # at the end of the command will not affect the execution of the command since anything that follows # is treated as a comment in Bash—the text is silently ignored after #.

Let us understand the above procedure using an example.

Let us assume that I run the following command on my Bash shell:

```
$ tar xzvf filename #untar
```

Now, just press *Ctrl + R* in the terminal window and type *untar*; you will get the above command from the Bash history of commands.

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Delete files that are x days old

Sometimes in Linux, you want to clear out older files in a directory. One example could be if you have a security system and it continuously writes video files to a directory on your NAS (Network Attached Storage) until it fills it up. You've figured out that if you keep a week's collection of videos, there will be plenty of space for other users. So, here is a command that will delete all files that are older than seven days. Do remember to execute this command with caution as it can delete your important data if not used correctly:

```
#find /path/to/files/ -type f -mtime +7 -exec rm -rf {} \;
```

Let us look at the above command more closely.

find: This is the command that will search for the files.

/path/to/files/: This is the top level directory to start searching.

-type f: This ensures we don't remove directories, but only files.

-mtime +7: Removes files older than '7' days. Change to '+14' to delete files older than two weeks.

-exec: This indicates what to do with the files we found.

rm -rf: Removes the files recursively and forcefully.

{}: This represents each file we find.

\;: This is the end of the exec.

On successfully testing the above command you can create a *cron* job to automate the process. The *cron* job entry given below in the *crontab* file will execute the above command every night at 2 a.m. and delete all files older than 7 days in the folder.

```
2 * * * /bin/find /path/to/files/ -type f -mtime +7 -exec rm -rf {} \;
```

—Ninad Shaha, ninadshaha@iitb.ac.in



How to resize a Linux partition

This tip will help you to resize your partition easily in a few steps, without the need to reinstall the operating system again.



Note: It is recommended to take backups of all the files.

Open up a terminal and run the following command:

```
$ sudo init 1
```

The '*init*' command will load *runlevel 1*, which is used to carry out administrative operations like maintenance or emergency repairs.

Here, we are going to extend the root partition using free space from the home partition to make space for data in the root folder. The partitions should be a logical volume (which is more flexible than other partition schemes) and of file system type *ext3/ext4*.

Inside *runlevel 1* run the following commands:

```
$ cd /
```

```
$ umount /home
```

Here, we are changing our directory to root so that we can unmount */home* and modify the partition's size.